Finding Thurstone: modeling comparative judgment data with `R` (and `Stan`)

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Abstract

The Classical Comparative Judgment (CCJ) analysis has become the standard approach for analyzing comparative judgment (CJ) data across multiple fields. Researchers favor CCJ because it offers a relatively simple approach for trait measurement and follow-up analyses. This simplicity stems from three features: its reliance on restricted Thurstone-based models to estimate latent traits, its limited consideration of relevant assessment design features, and its separation of trait measurement from follow-up analysis. Recent studies, however, question whether these modeling assumptions adequately reflect contemporary CJ assessments and whether omitting relevant design features, together with the separation of measurement and analysis, may bias estimates and reduce inferential precision.

To address these concerns, Rivera et al. (2025) proposed the Information-Theoretical Comparative Judgment (ITCJ) analysis, an umbrella analytical workflow designed to accommodate the complexity observed in contemporary CJ assessments. The approach enables researchers to specify models tailored to the assumed CJ data-generating process, thereby reducing reliance on simplifying assumptions. In addition, by integrating measurement, analysis, and relevant assessment design features within a single framework, the approach also supports coherent estimation and inference. However, these potential advantages require empirical validation.

Therefore, this study evaluates the ITCJ analysis using a synthetic speech-quality dataset and benchmarks its performance against the CCJ approach. In line with principles of open and reproducible science, the study also provides step-by-step guidance on data simulation, prior specification, model estimation, and interpretation using `R`, `Stan`, `cmdstan`, and `brms`. Ultimately, by combining empirical validation with a fully reproducible implementation, the study equips researchers with practical tools for applying the ITCJ approach to complex CJ assessments.

Keywords: tutorial, causal inference, bayesian inference, thustone, comparative judgement

1. Introduction

Comparative judgment (CJ) has emerged as a valuable methodology for measuring latent traits across diverse fields, including education (Kimbell, 2012; Jones and Inglis, 2015; van Daal et al., 2016; Bartholomew et al., 2018), psychology (Maydeu-Olivares and Böckenholt, 2005), political sciences (ZuccoJr. et al., 2019), linguistics (Boonen et al., 2020), and criminology (Seymour et al., 2022; Seymour and Hernandez, 2025). In CJ studies, judges actively compare pairs of stimuli to determine which stimulus exhibits more of the latent trait of interest (Thurstone, 1927a,b).

The *Classical Comparative Judgment* (CCJ) analysis has become the standard approach for analyzing CJ data across multiple fields, including education, linguistics, and psychology (see, for example, Wu, 2025; Thwaites and Paquot, 2024; Maydeu-Olivares and Böckenholt, 2005). Researchers favor CCJ because it provides a relatively simple method for measuring traits and conducting follow-up analyses (Andrich, 1978; Pollitt, 2012b). Three features underpin its simplicity. First, it relies on restricted Thurstone-based models to estimate latent traits. These models facilitate estimation by imposing simplifying assumptions about the traits, judges, and stimuli involved in CJ assessments (Thurstone, 1927b; Bramley, 2008). Second, it pays limited attention to relevant CJ assessment design features that shape the data-generating process, such as the sampling and comparison mechanisms and potential judge biases (Rivera et al., 2025). Third, the approach separates trait estimation from follow-up analysis (Pollitt, 2012b), simplifying the latter by focusing only on mean trait estimates or residuals obtained from the former process (Rivera et al., 2025).

Recent studies, however, question whether these modeling assumptions hold in contemporary CJ applications, and whether the limited consideration of relevant assessment design features, together with the separation of trait estimation and analysis, may bias estimates and reduce the precision of inferences. (Bramley, 2008; Kelly et al., 2022; Rivera et al., 2025).

To address these concerns, Rivera et al. (2025) proposed the *Information-Theoretical Comparative Judgment* (ITCJ) analysis, an umbrella analytical workflow designed to accommodate the complexity often observed in applied CJ settings. The approach leverages Causal and Bayesian inference methods to combine Thurstone's core theoretical principles with relevant CJ assessment design features. This integration enables researchers to develop models tailored to the assumed data-generating process of the CJ system under study, thereby reducing reliance on simplifying assumptions. Moreover, by integrating measurement, analysis, and relevant design features within a single framework, the ITCJ analysis facilitates a single, coherent and flexible approach for trait estimation and statistical inference.

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The flexibility of the ITCJ analysis is especially valuable in education, linguistics, and psychology, where CJ data often arise from complex assessment designs (Bramley, 2008; van Daal et al., 2017; Kelly et al., 2022) or involve comparisons of complex, heterogeneous stimuli (van Daal et al., 2016; Lesterhuis et al., 2018; Chambers and Cunningham, 2022). Its relevance, however, extends beyond these fields to any discipline using CJ assessments, as the approach allows researchers to model the processes of substantial interest rather than those constrained by restrictive statistical models.

1.1. Research goals

The ITCJ analysis holds theoretical promise for producing reliable trait estimates and accurate statistical inferences. However, as Rivera and colleagues (2025) acknowledge, these potential advantages remain empirically untested. This study addresses this gap by *empirically validating the ITCJ analysis* using a synthetic speech-quality dataset. Specifically, it evaluates the accuracy and reliability of the resulting trait estimates and inferential parameters obtained under the ITCJ approach relative to those produced by the standard CCJ analysis.

In addition, consistent with principles of open and reproducible science, the study also provides step-by-step guidance on data simulation, prior specification, model estimation, and interpretation using `R` (R Core Team, 2015), `Stan` (Stan Development Team., 2026b,a), and the interface packages `cmdstanr` (Gabry et al., 2025) and `brms` (Bürkner, 2017, 2018). Notably, by combining empirical validation with a fully reproducible implementation, the study equips researchers with practical tools needed to apply the ITCJ approach to more complex CJ assessments.

The remainder of this manuscript is organized into five sections. Section 2 reviews the standard analytical approach applied to CJ data: the CCJ analysis, and compares it with the ITCJ analysis. Section 3 describes the assumed data-generating process for the simulated dataset, the simulation procedure, the practical implementation of each analytical approach, and the evaluation criteria aligned with the research goals. Section 4 presents the data description and modeling results. Section 5 interprets the findings, outlines future research directions, and considers the study limitations. Finally, Section 6 offers the concluding remarks.

2. A tale of two analytical approaches

In education, linguistics, and psychology, researchers typically analyze pairwise comparison data using the standard *CCJ analysis*. However, Rivera and colleagues (2025) propose an alternative approach, the *ITCJ analysis*. This section provides an overview of both methods.

2.1. The CCJ analysis

The CCJ analysis typically proceeds through two sequential steps, each serving a distinct purpose: first, estimating traits, and second, conducting follow-up analyses (Pollitt, 2012a,b; Jones et al., 2019; Boonen et al., 2020; Chambers and Cunningham, 2022; Bouwer et al., 2023). Trait estimation generally relies on predefined Thurstone-based models, such as the Bradley-Terry-Luce (BTL) model (Bradley and Terry, 1952; Luce, 1959), to generate two primary outputs: mean trait estimates and comparison-level residuals. For example, Boonen et al. (2020) applied the BTL model to derive mean latent scores representing children's overall speech quality, together with the corresponding comparison-level residuals.

Researchers then use these mean trait estimates and comparison-level residuals to conduct follow-up analyses. On the one hand, analyses of the mean latent scores fulfill several purposes, including summarizing estimates to aid interpretation, aggregating stimulus-level estimates to the individual level, partitioning variability between and within individuals, and drawing inferences about factors that influence trait scores. For example, Maydeu-Olivares and Böckenholt (2005) applied various factor models to latent means that represent individuals' preferences for compact cars, to provide a more parsimonious representation of these preferences. Similarly, Boonen et al. (2020) applied a multilevel regression model to mean latent estimates to examine whether children's age or hearing status affects their speech quality scores.

On the other hand, analyzing the residuals typically seeks to aggregate the remaining variability at the judge level, partition residual variability between and within judges, test for systematic biases, and identify potential misfitting stimuli, individuals, or judges. For instance, Wu (2025) fitted an analysis of variance (ANOVA) model to judges' infit statistics to examine whether judge expertise influences judgment behavior. The infit statistics–a weighted average of the squared Pearson residuals–indicate the extent to which judges' decisions deviate from expectations relative to those of other judges (Wright and Masters, 1982).

This two-step CCJ analysis has become a standard practice in many CJ applications across education, linguistics, and psychology (see, e.g., Wu, 2025; Thwaites and Paquot, 2024; Maydeu-Olivares and Böckenholt, 2005). However, Rivera et al. (2025) argued that this approach has three key limitations: first, it relies on restrictive Thurstone-based models for trait estimation; second, it overlooks relevant CJ design features; and third, it separates trait estimation from subsequent analyses.

Regarding the first limitation, although restricted Thurstone-based models facilitate trait estimation by imposing simplifying assumptions about judges, individuals, and stimuli (Thurstone, 1927b; Bramley, 2008), violations of these assumptions can undermine the reliability and validity of the resulting trait estimates (Perron and Gillespie, 2015; Rivera et al., 2025). This concern is exemplified by Wu (2025), who, under the BTL model, implicitly treats second-language English texts as independent stimuli that are equally informative about individuals' writing performance, despite their heterogeneous and complex nature (Thurstone, 1927a; Andrich, 1978; Bramley, 2008; Kelly et al., 2022). Notably, if the texts differ in informativeness due to this heterogeneous nature, the BTL assumption of equal dispersions may be violated, potentially leading to an overestimation of the trait reliability and misleading conclusions about differences between stimuli (McElreath, 2020; Wu et al., 2022). Moreover, if the characteristics of these texts unevenly influence judges' perceptions, systematic biases may arise. Such biases can induce dependencies among stimuli, further contributing to an over- or underestimation of the trait reliability (van der Linden, 2017; Ackerman, 1989; Hoyle, 2023).

Concerning the second limitation, while downplaying specific CJ assessment design features may also simplify trait estimation or follow-up analyses, Rivera and colleagues (2025) argued that this comes at a cost. In particular, ignoring elements that shape the CJ data-generating process can omit important latent dimensions, such as judge bias, or obscure the role of core design components like the sampling and comparison mechanisms. Crucially, neglecting judge bias can produce biased trait estimates and residuals (Reckase et al., 1986; Ackerman, 1989; Kahraman, 2013; Hoyle, 2023), while ignoring sampling and comparison mechanisms can distort both estimation and statistical inference (Kohler et al., 2019; McElreath, 2020, chap. 15; Schuessler and Selb, 2023).

Lastly, concerning the third limitation, although separating trait estimation from follow-up analysis can simplify the latter, focusing solely on mean trait estimates or residuals obtained from the estimation stage inadvertently treats these outputs as fixed values rather than as uncertain parameter estimates (McElreath, 2020; Rivera et al., 2025). This separation prevents the propagation of uncertainty from the estimation to the analysis phase (Pritikin, 2020), which can bias parameter estimates and reduce their precision. The direction and magnitude of these biases are often difficult to predict. Effects may be attenuated, inflated, or remain unchanged depending on the level of uncertainty in the scores and the true effects under investigation (McElreath, 2020; Kline, 2023; Hoyle, 2023). In addition, reduced precision lowers statistical power and increases the risk of Type I and Type II errors when conducting statistical inference (McElreath, 2020).

Several studies have made valuable attempts to incorporate specific features that address some of the limitations discussed above. However, most studies propose incremental models that address one or two issues at a time. As a result, other relevant components of the CJ process remain unrepresented, which may lead to model misspecification and produce biased or imprecise parameter estimates (Everitt and Skrondal, 2010). For example, Seymour et al. (2022) and Seymour and Hernandez (2025) extended the BTL model to incorporate spatially correlated stimuli and tied outcomes, applying these models to measure area-level deprivation and honour-based abuse, respectively. Although these extensions captured important aspects of the CJ data, the analysis did not explicitly model other potentially relevant factors, such as the sampling mechanism associated with their "convenience" data sample, possible judge biases and judge-specific effects, or hierarchical structures expected in the data, such as wards nested within counties. Similar patterns of underrepresented features appear in Pritikin (2020), Schramm and Batchelder (2021), and Gray et al. (2024), where models typically address one or two key design features or sources of heterogeneity at a time.

Notably, the *ITCJ analysis* proposed by Rivera and colleagues (2025) offers a systematic and integrated approach that addresses all the limitations outlined above.

2.2. The ITCJ analysis

In contrast to the CCJ approach, the ITCJ analysis does not begin by applying a predefined statistical model to CJ data and then extracting outputs for subsequent analysis. Instead, it begins by explicitly representing the relationships among observed and latent variables that are thought to characterize the data-generating process of the CJ system under study. The analysis formalizes these relationships through *Directed Acyclic Graphs (DAGs)* and an associated *Structural Causal Models (SCMs)* (Morgan and Winship, 2014; Gross et al., 2018; Neal, 2020). It then derives a *probabilistic model* from this causal representation and formulates one or more *statistical models* tailored to the system under study. Finally, the approach uses these models to estimate, summarize, and draw statistical inferences about the traits of interest.

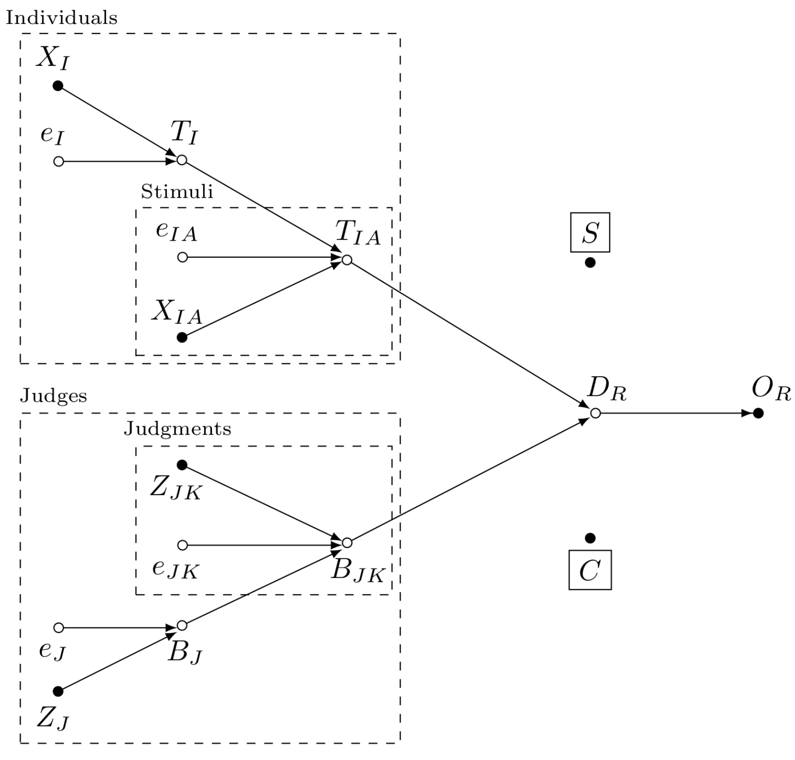
As an illustration, Figure 1 presents a plausible data-generating process for a CJ-based speech-quality assessment. The SCM in Figure 1a formally describes the non-parametric relationships among observed and latent variables through structural equations, denoted by $f_{(\cdot)}$. The SCM uses the symbol ':=' to denote the asymmetric direction of causal influence and the symbol ' $\perp$ ' to represent d-separation, a concept akin to conditional independence (Pearl, 2009; Cinelli et al., 2020). In contrast, the DAG in Figure 1b represents the same structure in graphical form, where observed variables appear as solid circles and latent (unobserved) variables as open circles. Arrows indicate the direction of the assumed causal influences, and dashed boxes represent the hierarchical structure of the data, such as stimuli nested within individuals. Rivera et al. (2025) provides a more detailed discussion of these concepts in the context of CJ assessments.

Specifically, Figure 1b depicts the assumption that individual-level variables X_I causally influence individual traits T_I, i.e., the individuals' ability to produce high-quality speech. Examples of individual-level variables include hearing status and age, the latter serving as a proxy for language development. The DAG also assumes that stimulus-level variables X_{IA}, together with individual traits T_I, causally influence stimulus traits T_{IA}, which represent the latent quality of the stimulus under comparison. Examples of such variables include speech-sample length and whether the sample originates from spontaneous speech or imitation. The SCM in Figure 1a formalizes these relationships through the structural equations $T_I := f_T(X_I, e_I)$ and $T_{IA} := f_T(T_I, X_{IA}, e_{IA})$, where the error terms e_I and e_{IA} capture residual variation at the individual and stimuli levels, respectively.

Furthermore, because the stimulus traits T_{IA} are only indirectly accessible through judges' perceptions, and these perceptions may systematically deviate from consensus (see, Pollitt and Elliott, 2003;

$$
\begin{aligned}
O_R &:= f_O(D_R, S, C) \\
D_R &:= f_D(T_{IA}, B_{JK}) \\
T_{IA} &:= f_T(T_I, X_{IA}, e_{IA}) \\
T_I &:= f_T(X_I, e_I) \\
B_{JK} &:= f_B(B_J, Z_{JK}, e_{JK}) \\
B_J &:= f_B(Z_J, e_J) \\
e_I &\perp \{e_J, e_{IA}, e_{JK}\} \\
e_J &\perp \{e_{IA}, e_{JK}\} \\
e_{IA} &\perp e_{JK}
\end{aligned}
$$

(a) SCM



(b) DAG

Figure 1: ITCJ analysis, plausible data-generating process for a CJ-based speech quality study.

van Daal et al., 2016), the SCM and DAG treat judgment-level biases as an integral component of the CJ system from the outset (Rivera et al., 2025). Accordingly, Figure 1b depicts the assumption that judge-level variables Z_J causally influence judge-level bias B_J, that is, the systematic deviation of a judge's perception of a stimulus from the consensus. Examples of judge-level variables include speech-judgment expertise and prior training in the CJ task. Moreover, the DAG assumes that judgment-level variables Z_{JK}, together with judge-level biases B_J, causally influence judgment-level biases B_{JK}, which represent systematic deviations of an individual judgment from the judge-specific bias. Examples of such variables include the time of day or day of the week at which the comparison was made, as well as the number of comparisons completed by a judge, used as a proxy for learning or fatigue effects during the CJ task. As before, the SCM in Figure 1a formalizes these relationships through the structural equations $B_J := f_B(Z_J, e_J)$ and $B_{JK} := f_B(B_J, Z_{JK}, e_{JK})$, where the error terms e_J and e_{JK} capture residual variation at the judge and judgment levels, respectively.

Figure 1 also encodes two additional assumptions. First, it specifies that the *discriminal difference* D_R depends on both the relative distance between stimulus traits and judgment-level biases (Rivera et al., 2025). Second, it specifies that the observed outcome O_R depends on the discriminal difference (Thurstone, 1927a,b). The DAG represents these dependencies with arrows from T_{IA} and B_{JK} to D_R, and from D_R to O_R, while the SCM formalizes the same relationships through the structural equations $D_R := f_D(T_{IA}, B_{JK})$ and $O_R := f_O(D_R, S, C)$.

Finally, Figure 1 explicitly represents the sampling and comparison mechanisms, denoted by S and C, respectively. Following Rivera et al. (2025), the SCM and DAG define both mechanisms as binary vectors. Specifically, $S = 1$ indicates that the corresponding individual, stimulus, and judge from the conceptual population are included in the sample, whereas $C = 1$ indicates that the corresponding pairwise comparison from the conceptual population is included in the observed data. Together, these mechanisms determine how conceptual-population outcomes become observed comparison outcomes.

In this context, the DAG in Figure 1b depicts S and C without incoming arrows and encloses both variables in squares, reflecting two design assumptions. First, the absence of incoming arrows indicates that both variables follow random design mechanisms. Specifically, S follows a *simple random sampling (SRSg)* design, in which each judgment, judge, stimulus, and individual has an equal probability of selection (Lawson, 2015). Likewise, C follows a *random allocation comparative design*, also known as an *incomplete block design* (Bramley, 2015; Lawson, 2015), in which each pairwise comparison has an equal probability of assignment to any judge.

Second, the squares surrounding S and C denote *conditioning* (Bareinboim et al., 2014; Schuessler and Selb, 2023). Informally, conditioning means that the current CJ structure applies only to observations satisfying $S = 1$ and $C = 1$ (Neal, 2020; McElreath, 2020), that is, the observed sample comparison data. Crucially, because the random mechanisms governing S and C satisfy ignorability (Schuessler and Selb, 2023) and related identification properties, the inferences drawn from this conditioned CJ structure remain valid for the conceptual population (Morgan and Winship, 2014; Neal, 2020). For an introduction to identification and ignorability in CJ assessments, see Rivera et al. (2025).

$$
\begin{aligned}
P(O_R, D_R, S, C, T_{IA}, X_{IA}, e_{IA}, T_I, X_I, e_I, B_{JK}, Z_{JK}, e_{JK}, B_J, Z_J, e_J) \\
= P(O_R \mid D_R, S, C) \cdot P(S) \cdot P(C) \\
\cdot P(D_R \mid T_{IA}, B_{JK}) \\
\cdot P(T_{IA} \mid T_I, X_{IA}, e_{IA}) \cdot P(X_{IA}) \\
\cdot P(T_I \mid X_I, e_I) \cdot P(X_I) \\
\cdot P(B_{JK} \mid B_J, Z_{JK}, e_{JK}) \cdot P(Z_{JK}) \\
\cdot P(B_J \mid Z_J, e_J) \cdot P(Z_J) \\
\cdot P(e_I) P(e_{IA}) P(e_J) P(e_{JK})
\end{aligned}
$$

Figure 2: Probabilistic model, joint distribution (left hand side) factorized into conditional probability distributions (CPDs, right hand side).

Once researchers specify the assumptions characterizing the CJ data-generating process in the form of an SCM, as in Figure 1a, they can translate it into a probabilistic model. This model factorizes the joint distribution of the complex CJ system into a product of simpler *conditional probability distributions (CPDs)* (Pearl et al., 2016; Neal, 2020), as illustrated in Figure 2. From this formulation, researchers can then derive one or more statistical models tailored to the CJ system under study (see Rivera et al., 2025). This derivation enables coherent trait estimation and inference within a unified framework.

At this point, it becomes clear that the ITCJ approach differs fundamentally from the CCJ approach in how it conducts trait estimation and inference. First, although the ITCJ analysis still relies on Thurstone-based models for trait estimation, these models are no longer treated as pre-specified simplifying abstractions; instead, they are adapted to the specific data-generating process of the CJ system under study, enabling researchers to explicitly model the processes of substantive interest. Second, the approach explicitly incorporates relevant CJ assessment design features, including the sampling and comparison mechanisms as well as judge- and judgment-level biases. Third, rather than separating trait estimation from follow-up analyses, the ITCJ analysis integrates both within a unified and systematic framework, enabling the propagation of uncertainty across all levels of the assumed hierarchical data structure (Rivera et al., 2025). Ultimately, these innovations allow the ITCJ analysis to address the three main limitations of the CCJ approach.

Building on this framework, the following section details the methods used to implement and evaluate the ITCJ analysis against the CCJ approach.

3. Methods

To address its goal of empirical model validation using a synthetic speech-quality dataset, this study follows the *Bayesian research workflow* (Depaoli and van de Schoot, 2017; Pearl and Mackenzie, 2018; Neal, 2020; Gelman et al., 2020; Schad et al., 2020; Betancourt, 2020; McElreath, 2024a,b), depicted in Figure 3. This workflow guides the construction of the synthetic dataset, as well as the specification, estimation, and evaluation of the analysis models.

Notably, as shown in the figure, the workflow components form an interconnected and partly non-linear process, reflecting the inherent complexity of (Bayesian) statistical modeling. To ensure clarity and reproducibility, the remainder of this article unpacks each stage step-by-step.

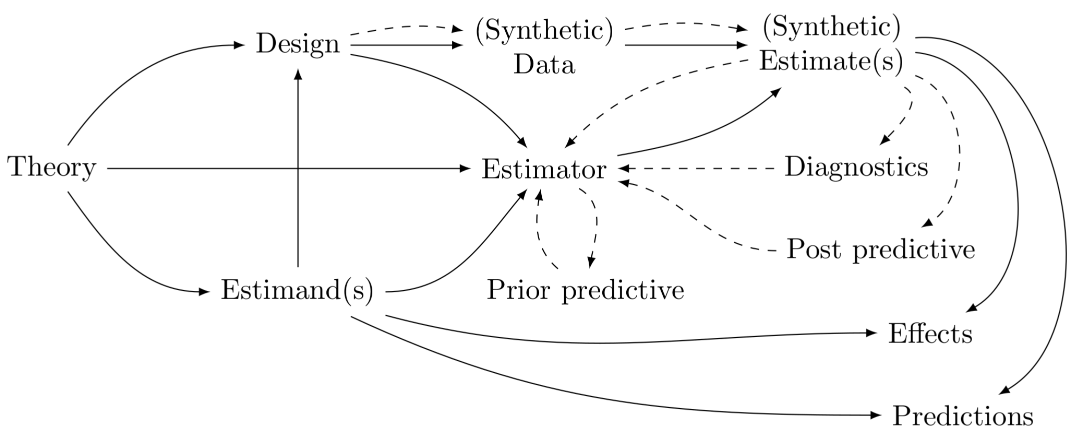


Figure 3: Bayesian research workflow.

3.1. From Theory to Design: Data-generating assumptions

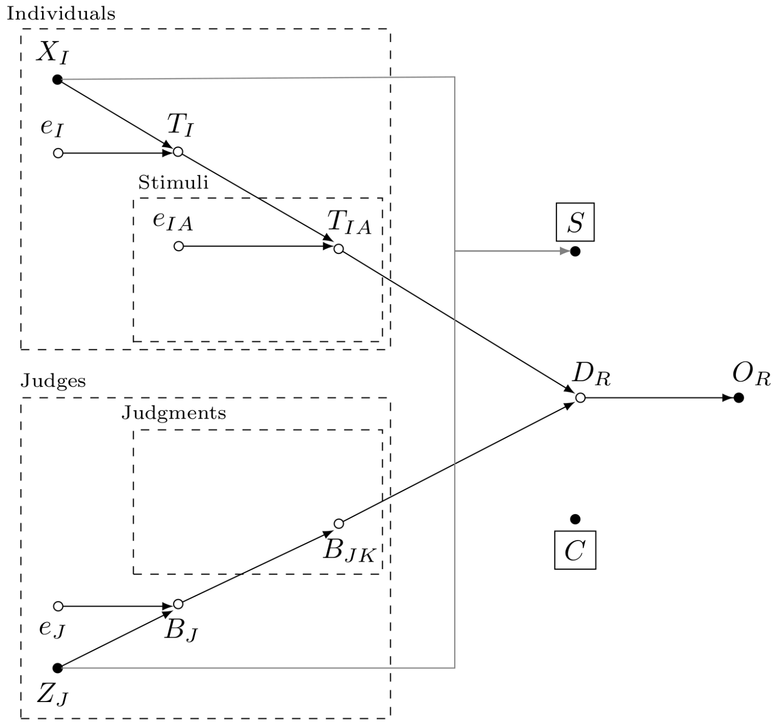
Because this study aims to conduct empirical validation, it employs a synthetic dataset. Synthetic data enables rigorous evaluation of model performance under controlled conditions, where both the underlying data-generating mechanisms and the target parameters are known by design.

The research question and data-generating assumptions of the synthetic dataset used in this study draw on key design features and findings reported in Boonen et al. (2020). The research question retains the original study's objective of examining whether listeners can perceive differences in the overall speech quality of children. Similarly, the data-generating assumptions preserve the core design features of the original CJ task, in which three groups of listeners with different levels of experience with the speech of hearing-impaired (HI) children–namely, audiologists (AU), primary school teachers (PT), and inexperienced listeners (IL)–compare pairs of stimuli and judge which sounds better.

The stimuli consist of short speech samples produced by children from three hearing-status groups: normal-hearing (NH), hearing aid users (HI-HA), and cochlear implant users (HI-CI).

$$
\begin{aligned}
O_R &:= f_O(D_R, S, C) \\
D_R &:= f_D(T_{IA}, B_{JK}) \\
T_{IA} &:= f_T(T_I, e_{IA}) \\
T_I &:= f_T(X_I, e_I) \\
B_{JK} &:= f_B(B_J) \\
B_J &:= f_B(Z_J, e_J) \\
S &:= f_S(X_I, Z_J) \\
e_I &\perp \{e_J, e_{IA}\} \\
e_J &\perp \{e_{IA}\}
\end{aligned}
$$

(a) SCM



(b) DAG

Figure 4: Assumed data-generating process underlying the synthetic conceptual population

Figure 4 presents the SCM and DAG incorporating these design features. The DAG in Figure 4b represents children's characteristics X_I as causally influencing their speech quality trait T_I. Consistent with the original study, the relevant child characteristics include age (XIc) and the three aforementioned hearing status groups: NH ($XId[1]$), HI-HA ($XId[2]$), and HI-CI ($XId[3]$) children. Moreover, the DAG depicts children's speech quality trait T_I as causally influencing the stimulus traits T_{IA}, while excluding additional stimulus-level attributes X_{IA}. The SCM in Figure 4a formalizes these relationships through the structural equations $T_I := f_T(X_I, e_I)$ and $T_{IA} := f_T(T_I, e_{IA})$, where the error terms e_I and e_{IA} capture residual variation at the children and stimuli levels, respectively.

Likewise, the Figure 4b depicts judge-level characteristics Z_J as causally influencing judge-level biases B_J, where the error term e_J captures residual variation of the bias at the judge level. Consistent with the original study, the relevant judge-level characteristic is expertise in speech-quality assessment, represented by the three aforementioned groups: AU ($ZJd[1]$), PT ($ZJd[2]$), and IL ($ZJd[3]$). The graph also assumes that each judge performs a single judgment per stimulus pair, which explains the absence of variables Z_{JK} and e_{JK} previously shown in Figure 1b. The SCM in Figure 4a formalizes these relationships through the structural equations $B_J := f_B(X_I, e_I)$ and $B_{JK} := f_B(B_J)$.

Finally, the DAG in Figure 4b represents the sampling and comparison mechanisms, S and C, respectively. Following the original study and consistent with Figure 1b, the graph assumes that the comparison mechanism C follows a random allocation comparative design (Bramley, 2015), as indicated by the absence of incoming arrows. The sampling mechanism S, however, follows a *randomized blocked* design (Lawson, 2015), as indicated by the incoming arrows. In these designs, the probability of selection for children and judges depends on their characteristics. This assumption departs from the corresponding assumption in Figure 1b and requires the additional structural equation $S := f_S(X_I, Z_J)$. The implications of this assumption for the simulation and analysis of the synthetic data are discussed in Section 3.2, Section 3.3, and Section 3.4.

TO DO

$$O_R := f_O(D_R, S, C)$$
$$D_R := f_D(T_{IA}, B_{JK})$$
$$T_{IA} := f_T(T_I, e_{IA})$$
$$T_I := f_T(X_I, e_I)$$
$$B_J := f_B(Z_J, e_J)$$

$$e_{IA} \perp \{e_I, e_J\}$$
$$e_I \perp \{e_J\}$$

(a) SCM

$$P(O_R \mid D_R, S, C)$$
$$P(D_R \mid T_{IA}, B_{JK})$$
$$P(T_{IA} \mid T_I, e_{IA})$$
$$P(T_I \mid X_I, e_I)$$
$$P(B_J \mid Z_J, e_J)$$

$$P(e_{IA})P(e_I)P(e_J)$$

(b) Probabilistic model

$$O_R \overset{iid}{\sim} \text{Bernoulli}\left[\text{inv\_logit}(D_R)\right]$$
$$D_R = (T_{IA}[i,a] - T_{IA}[h,b]) + B_{JK}[j,k]$$
$$T_{IA} = T_I + e_{IA}$$
$$T_I = \beta_{XI} X_I + e_I$$
$$B_J = \beta_{ZJ} Z_J + e_J$$

$$e \sim \text{Multi-Normal}(\mu, \Sigma)$$
$$\Sigma = VQV$$

(c) Statistical model

Figure 5: SCM, Probabilistic and Statistical model of the "True" Data-Generating Process Underlying the Synthetic Conceptual Population

Next, the study incorporates Boonen and colleagues' (2020) empirical findings by specifying a statistical model derived from the probabilistic model implied by the system's SCM. This procedure is depicted in Figure 5. For this purpose, the model first organizes individual-level predictors into the vector $X_I = \{XIc, XId[1], XId[2], XId[3]\}$ with corresponding effect vector $\beta_{XI} = \{\beta_{XIc}, \beta_{XId[1]}, \beta_{XId[2]}, \beta_{XId[3]}\}$, and judge-level variables into the vector $Z_J = \{ZJd[1], ZJd[2], ZJd[3]\}$ with corresponding effect vector $\beta_{ZJ} = \{\beta_{ZJd[1]}, \beta_{ZJd[2]}, \beta_{ZJd[3]}\}$.

The model then assumes that each one-unit increase in age (XIc) leads to a 0.1-logit increase in speech quality for children aged five to nine, as captured by $\beta_{XIc} = 0.1$. It further encodes a strict ordering of effects across three hearing-status groups, such that normal-hearing children ($NH; XId[1]$) exhibit higher speech quality than hearing-impaired children with hearing aids ($HI - HA; XId[2]$), who in turn outperform hearing-impaired children with cochlear implants ($HI - CI; XId[3]$). This ordering is represented by $\beta_{XId[1]} > \beta_{XId[2]} > \beta_{XId[3]}$, where $\beta_{XId[1]} = 1$, $\beta_{XId[2]} = 0$, and $\beta_{XId[3]} = -1$, corresponding to a one-logit difference between adjacent groups.

Moreover, the model assumes that expertise with the judgment task does not systematically influence average judgment bias. Consequently, audiologists ($AU; ZJd[1]$), primary teachers ($PT; ZJd[2]$), and inexperienced listeners ($IL; ZJd[3]$) all have zero mean effects, such that $\beta_{ZJd[1]} = \beta_{ZJd[2]} = \beta_{ZJd[3]} = 0$.

Similarly, to incorporate assumptions about the covariance structure of the error terms, the model collects them in the vector $e = \{e_I, e_{IA}, e_J\}$. It then assumes that this vector follows a Multivariate Normal distribution with mean $\mu = [0, 0, 0]^T$ and covariance matrix Σ. Moreover, it specifies the covariance matrix via a *standard deviation–correlation decomposition* (McElreath, 2020), such that

$\Sigma = VQV$, where V is a diagonal matrix of standard deviations and Q is a correlation matrix. More specifically, the study defines the matrices Q and V as follows:

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; V = \begin{bmatrix} s_{XI} & 0 & 0 \\ 0 & s_A & 0 \\ 0 & 0 & s_{ZJ} \end{bmatrix}$$

Figure 6: Assumptions for the standard deviation-correlation decomposition of the covariance matrix Σ.

The Q matrix indicates that the generating model assumes the three types of error terms are mutually independent. This implies that residual variability in stimulus traits is uncorrelated with residual variability in individual traits, and that both are uncorrelated with residual variability in judges' biases.

Moreover, the model encodes additional assumptions through the V matrix. Specifically, it assumes that variability in individuals' ability to produce high-quality speech differs across hearing-status groups. In particular, NH children exhibit higher variability ($s_{XId[1]} = 1.5$) than $HI-HA$ and $HI-CI$ children, both of whom are assumed to show lower and equal variability ($s_{XId[2]} = s_{XId[3]} = 0.75$). Considering these specifications, and assuming an equal-probability SRSg design, the model implies an average between-individual trait variability of $s_{XI} = \sum_{g=1}^{3} s_{XId[g]}/3 = 1$, which exceeds the assumed within-individual (stimulus-level) trait variability $s_A = 0.2$.

Finally, although the original study found no evidence that judges' expertise influences biases in their speech-quality perceptions, it did not examine whether expertise affects the *variability* of those biases. To assess the ability of the ITCJ approach to investigate such hypothesis, the generating model assumes that audiologists (AU) exhibit less bias variability than primary teachers (PT), who in turn exhibit less bias variability than inexperienced listeners (IL). This ordering is represented by $s_{ZJd[1]} < s_{ZJd[2]} < s_{ZJd[3]}$, where $s_{ZJd[1]} = 0.5$, $s_{ZJd[2]} = 1$, and $s_{ZJd[3]} = 1.5$. Given these specifications, and an equal-probability SRSg design, the model implies an average between-judge bias variability of $s_{ZJ} = \sum_{g=1}^{3} s_{ZJd[g]}/3 = 1$.

3.2. From Design to Data: Data simulation

3.3. From Theory to Estimands: Target parameters

3.4. From Estimator to Estimate(s): The models

3.4.1. The CCJ analysis

3.4.2. The ITCJ analysis

3.4.2.1. Model 1.

3.4.2.2. Model 2.

3.4.2.3. Model 3.

3.4.2.4. Model 4.

3.4.2.5. Model 5.

3.4.2.6. Model 6.

3.5. From Estimate(s) to Diagnostics and Posterior predictives: The evaluation criteria

4. Results

4.1. Data description

4.2. Data modeling

4.2.1. The CCJ analysis

4.2.2. The ITCJ analysis

4.2.2.1. Model 1.

4.2.2.2. Model 2.

4.2.2.3. Model 3.

4.2.2.4. Model 4.

4.2.2.5. Model 5.

4.2.2.6. Model 6.

4.2.2.7. Model comparison.

5. Discussion

5.1. Future research directions

5.2. Study limitations

6. Conclusion

Declarations

Funding: The Research Fund (BOF) of the University of Antwerp funded this project.

Financial interests: The authors declare no relevant financial interests.

Non-financial interests: The authors declare no relevant non-financial interests.

Ethics approval: The University of Antwerp Research Ethics Committee confirmed that this study does not require ethical approval.

Consent to participate: Not applicable

Consent for publication: All authors have read and approved the final version of the manuscript for publication.

Data, materials and code availability: A previous version of this manuscript, along with the associated data, materials and code (see the section titled `CODE LINK`), has been made publicly available at: https://jriveraespejo.github.io/paper3_manuscript/.

AI-assisted technologies in the writing process: The authors used various AI-based language tools to refine phrasing, optimize wording, and enhance clarity and coherence throughout the manuscript. They take full responsibility for the final content of the publication.

CRediT authorship contribution statement: *Conceptualization:* J.M.R.E, T.vD., S.DM., and S.G.; *Methodology:* J.M.R.E, T.vD., and S.DM.; *Software:* J.M.R.E.; *Validation:* J.M.R.E.; *Formal Analysis:* J.M.R.E.; *Investigation:* J.M.R.E; *Resources:* T.vD. and S.DM.; *Data curation:* J.M.R.E.; *Writing - original draft:* J.M.R.E.; *Writing - review and editing:* J.M.R.E., T.vD., S.DM., and S.G.; *Visualization:* J.M.R.E.; *Supervision:* S.G. and S.DM.; *Project administration:* S.G. and S.DM.; *Funding acquisition:* S.G. and S.DM.

7. Appendix

7.1. Appendix A: Stationarity, converge and mixing

7.2. Appendix B: Misfit observations

7.3. Appendix C: Sample size calculations

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